

Employee Promotion Prediction Using Classification Models: SVM, Logistic Regression and Random Forest

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Abstract — Promotions can have a significant effect on employee motivation and the length of time an employee decides to stay in a company, while also impacting organizational effectiveness, since it can be more effective to promote skillful employees with strong expertise. This way, more work can be done cost-effectively. However, in many businesses, promotion decisions are often influenced by subjective evaluations. Having unclear criteria and process can lead to bias and misjudgment. This project explores the use of machine learning models to build a predictive system that identifies employees most likely to be promoted, based on HR data. Using a real-world dataset, we define the task as a binary classification problem, where the target variable `is_promoted` indicates whether an employee is promoted. We implemented three classification methods: Logistic Regression, Support Vector Machines (SVM), and Random Forest.

Our results showed that even simple classification models can produce reliable predictions. More importantly, this work emphasizes that machine learning can support fair and data-driven promotion decisions. Also By identifying key factors influencing promotions and predicting promotion outcomes reliably, organizations can enhance talent management, reduce bias and increase fairness.

Keywords — Support Vector Machine, Predictive models, Classification algorithms, employee promotion, machine learning, Logistic Regression, Random Forest

1. Introduction

In this project, we explore whether machine learning can support HR decision-making by predicting employee promotions using available HR data. Promotion prediction is still a relatively unexplored application of machine learning compared to other areas, such as hiring.

Recent studies in this area often propose complex or hybrid methods, such as combining Convolutional Neural Networks with Random Forest (Ardehkhani, Moslemi, & Hooshmand, 2023) or using boosting algorithms like CatBoost (Rallapalli & Kumar, 2024). While these approaches achieve strong performance, they can be challenging to interpret and apply in typical HR departments. Other researchers have applied classical models like Decision Trees, SVM, and Random Forest, often paired with resampling methods to address class imbalance (Şahinbaş, 2022; Al-Alawi & Albuainain, 2024).

Building on this classical approach, we emphasize practicality, interpretability, and ease of implementation. We use straightforward and reliable models (Logistic Regression, SVM, and Random Forest) and apply oversampling techniques to improve class imbalance. Our aim is to create an understandable and transparent framework that HR professionals can trust and implement in real-world settings.

2. Methods and Dataset

2.1. Dataset

We used an open-source Kaggle dataset (Arashnic Kaggle, 2020) from a large multinational company, containing 54,808

records with 13 features across nine departments, to identify employees eligible for promotion as well as having a Test dataset with 23,000+ records without the label. Table 1 is explaining the features of the dataset.

Table 1: data set features and Descriptions

Feature	Description
employee id	Unique ID for employee
department	Department of employee
region	Region of employment
education	Education Level
gender	Gender of Employee
recruitment channel	Channel of recruitment for employee
no of trainings	Number of other trainings completed
age	Age of Employee
previous year rating	Employee Rating for the previous
length of service	Length of service in years
awards won?	If awards won during previous year then 1 else 0
avg training score	Average score in current training evaluations
is promoted	(Target) Recommended for promotion

2.2. Preprocessing

Handling Missing Values:

Previous year rating: Filling missing values for this feature using the median, grouped by training score and awards won.

Education: Filling the missing values using Random Forest prediction model.

Removed the following irrelevant

Features:

employee_id and recruitment_channel.

Categorical Encoding: Applying one-hot encoding to department, region, and gender, with the first category dropped to avoid multicollinearity.

Scaling: Numeric features standardized using z-score normalization.

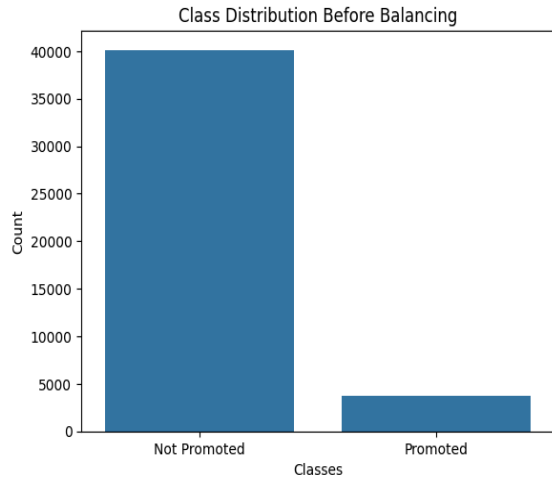


Figure 1: Class distribution

Handling Imbalanced Class distribution:

The dataset was imbalanced, with only about 8.5% of employees marked as promoted. This can cause models to favor the majority class and miss potential promotions. To fix this, we used two techniques: class weight adjustment and SMOTE. Class weighting adjusted the loss function by assigning higher weights to the minority class based on class frequencies.

The technique has been widely adopted for imbalanced classification tasks and was originally proposed by (Chawla et al , 2002). SMOTE creates new artificial examples for the minority class by combining features

from similar data points, helping the model better distinguish between classes without memorizing the training data. The validation and test sets were kept unchanged to make sure the evaluation was fair.

2.3. Methods

Model training:

We evaluated three supervised learning models Logistic Regression, SVM, and Random Forest for predicting employee promotions. To handle class imbalance, we applied SMOTE and class weighting. Logistic Regression and SVM were chosen for their interpretability, and we adjusted classification thresholds to optimize their performance. Random Forest served as a benchmark model, although it was less interpretable for practical HR use.

Threshold Tuning and Model Evaluation:

In imbalanced datasets, using the default classification threshold (0.5) often leads to poor performance for the minority class. In our case, false positives (promoting unqualified employees) carry a higher risk than false negatives. Threshold tuning is particularly important for imbalanced datasets, where maximizing the Area Under the Precision-Recall Curve is more informative than ROC-AUC. As noted by (Saito and Rehmsmeier, 2015). We optimized classification thresholds based on

the F1-score to balance precision and recall. Three classification models were evaluated using precision, recall, F1-score, ROC-AUC, and confusion matrices. Also, we tested the models on an unlabeled test set, and after retraining all three models after adding the predicted value for the test dataset, we saw almost the same result in the performance, indicating that model predictions are reliable.

After adding the threshold tuning, Logistic Regression (threshold=0.80) and SVM (threshold=0.52) showed slightly higher performance and interpretability compared to Random Forest, which was stable but less interpretable. Overall, threshold tuning slightly improved predictions, providing balanced results crucial for fair and practical HR decision-making.

3. Results and Discussion

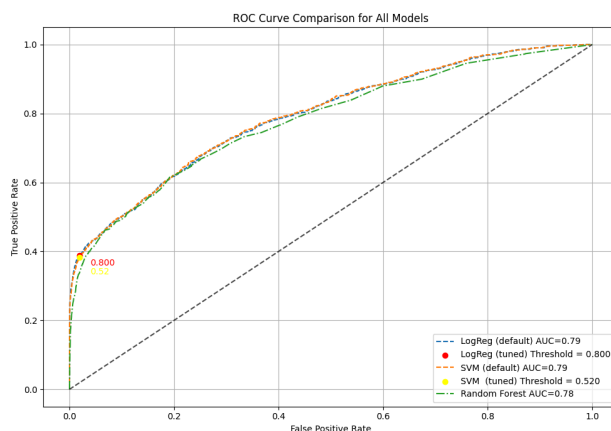


Figure 2: ROC Curve Comparison

Looking at feature importance across the models, we see that:

- **Performance score and recognition (awards won)** impact promotion predictions across all models.
- **Department** is important in linear models like Logistic Regression and SVM, but Random Forest relies on it less when other stronger features are available.
- **Length of service** helps in predicting promotions, but **age alone** does not impact promotion.

From the feature analysis, it's clear that previous year rating, average training score, awards won, and length of service are the factors with most impact in predicting promotion outcomes. All three models (Logistic Regression, SVM, and Random Forest) were able to effectively learn from the HR dataset and detect patterns linked to employee promotion. Although these models are relatively simple compared to more complex ensembles, they performed well and offered interpretability, which is especially important in HR contexts. In terms of metrics:

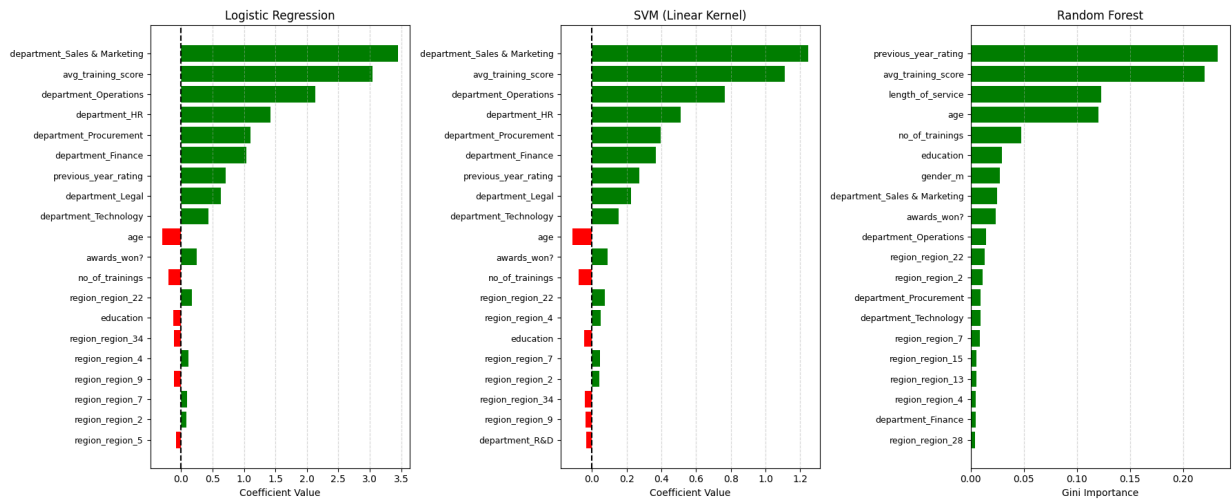


Figure 3: Top 20 Feature Importance

- Logistic Regression and SVM slightly outperformed Random Forest, by having higher AUC scores.
- Adjusting classification thresholds improved the balance between precision and recall, making the models better at identifying promoted employees without generating too many false positives.

Table 2: Model Evaluation Metrics

Model	ROC Score	Accuracy Score	F1 Score
Logistic Regression	0.7922	0.935	0.474
SVM	0.7932	0.934	0.369
Random Forest	0.7792	0.921	0.447

Also by using each model's own predictions as pseudo-labels on a second, unlabeled dataset and retraining, we saw that ROC and F1 either stay the same or improve slightly.

Showing that our models produce reliable predictions on new data.

4. Conclusion

This project used interpretable machine learning models (Logistic Regression, SVM, and Random Forest) to predict employee promotions using structured HR data. By addressing data imbalance and handling missing values, we built reliable models that support fairer and more consistent promotion decisions. Features such as average training score, previous year rating, and length of service proved to be strong predictors. While all models performed well, simpler models like Logistic Regression and SVM stood out for their interpretability and practical value in HR systems. Overall our approach offers a data-driven method for HR teams, with potential to scale.

References:

1. **Rallapalli, S., & Kumar, M. Y. M. (2024).** Optimizing employee promotion decisions: A novel machine learning framework for predictive analysis by using GBM CatBoost. In 2024 First International Conference on Software, Systems and Information Technology (SSITCON). IEEE. <https://doi.org/10.1109/SSITCON62437.2024.10796145>
2. **Ardehkhani, P., Moslemi, S. R., & Hooshmand, H. (2023).** Enhancing employee promotion prediction with a novel hybrid model integrating convolutional neural networks and random forest. In 2023 14th International Conference on Information and Knowledge Technology (IKT) (pp. 62–68). IEEE. <https://doi.org/10.1109/IKT62039.2023.10433023>
3. **Şahinbaş, K. (2022).** Employee promotion prediction by using machine learning algorithms for imbalanced dataset. In 2022 2nd International Conference on Computing and Machine Intelligence (ICMI) (pp. 1–6). IEEE. <https://doi.org/10.1109/ICMI55296.2022.9873744>
4. **Al-Alawi, A. I., & Albuainain, M. S. (2024).** Machine learning in human resource analytics: Promotion classification using data balancing techniques. In 2024 ASU International Conference in Emerging Technologies for Sustainability and Intelligent Systems (ICETSYS) (pp. 302–306). IEEE. <https://doi.org/10.1109/ICETSYS61505.2024.10459566>
5. **Arashnic. (2020).** HR analytics: Employee promotion data [Data set]. From Kaggle. <https://www.kaggle.com/datasets/arashnic/hr-ana>
6. **V. Chawla, K. W. Bowyer, L. O. Hall and W. P. Kegelmeyer,** "SMOTE: Synthetic Minority Over-sampling Technique," Journal of Artificial Intelligence Research, vol. 16, p. 321–357, 2002.
7. **T. a. R. M. Saito,** "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," PLOS ONE, vol. 10, no. 3, p. e0118432, 2015.